17/09/2025

Task-5.1

Implement various searching and sorting Operations in python programming.

Aim: To implement various searching and sorting operations in python programming.

Algorithm:

1) Input Definition:
2) Define the function find-employee-by-id that takes two parameters.
   a) A list of dictionaries (employees), where each dictio-nary represents an employee.
   b) An integer (target_id) representing the employee ID.
3) Iterate through the list.
4) check for Matching ID:
   Within the loop, check if the id field of the current dictionary matches the target_id.
5) Return matching Record:
   If a match is found.
6) Handle No match

Program:

```
def find-employee-by-id
    for employee in employees:
        if employee['id']==targed-id:
        return employee
    return None.
    # test the function.
    employees= [
     {'id':1,'name':'Alice','department':'HR'},
     {'id':2, 'name':'Bob', 'department':'engineering'},
     {'id':3, 'name':'charlie', department':'sales'},
     ]
print(find-employee-by-id(employees,2))#
output:{'id',:2,'name':'Bob','department':
'Engineering'}
```

Output:

{'Id': 2, 'name': 'Bob', 'department': 'Engineering'}

Task-5.2

Aim: It is to implement a feature that sorts the student records by their scores.

Algorithm:

1) Initialization
2) outer loop:
3) Track swaps
4) Inner loop
5) Compare and Swap
6) Early termination.
7) Completion.

Program:

```
def bubble-sort-scores
    n= len
    for i in range(n):
    # Track if any swap is made in this pass
     swapped = False
     for j in range(0, n-i-1):
        if students[j]['score'] > students[j+1]['score']:
       #swap if the score of the current student is grea
        -ter
       students[j], students[j+1] = students[j+1], students[j]
       swapped = True
  #if no two elements were swapped.
   if not swapped:
   break
 #Example usage
  students = [
  {'name':'Alice','score':88},
  {'name':'Bob','score':95},
  {'name':'charlie','score':75},
  {'name':'Diana','score':85}]
  print("Before sorting:")
  for student in students:
    print(student)
    bubble-sort-scores(students)
```

Output:-

['name' : 'Alice' , 'score' : 88}

{'name' : 'Bob' , 'score' : 95}

{'name' : 'charlie' , 'score' : 75}

{'name' : 'Diana' , 'score' : 85}

```python
print ("\n After Sorting:")
for student in students:
    print (student)
```

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Result: Thus, the program for various searching and Sorting Operations is executed and verified successfully.